

OS Objectives/Operations

- Process Management Activities
- Memory Management
- Storage Management
- Mass-Storage Management
- Protection System

Process Management Activities

The operating system is responsible for the following activities in connection with process management:

- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanisms for process synchronization
- Providing mechanisms for process communication
- Providing mechanisms for deadlock handling

Memory Management

- To execute a program
 - All (or part) of the instructions must be in memory
 - All (or part) of the data that is needed by the program must be in memory.
- Memory management determines what is in memory and when
 - Optimizing CPU utilization and computer response to users

Memory Management

- Memory management activities
 - Keeping track of the parts of memory are currently being used and by whom
 - Deciding which processes (or parts thereof) and data to move into and out of memory
 - Allocating and deallocating memory space as needed

Storage Management

- OS provides uniform, logical view of information storage
 - Abstracts physical properties to logical storage unit - **file**
 - Each medium is controlled by device (i.e., disk drive, tape drive)
 - Varying properties include
 - access speed,
 - capacity,
 - data-transfer rate,
 - access method (sequential or random)

Storage Management

- File-System management
 - Files usually organized into directories
 - **Access control** on most systems to determine who can access what
 - OS activities include
 - **Creating and deleting files and directories**
 - **Primitives to manipulate files and directories**
 - **Mapping files onto secondary storage**
 - **Backup files onto stable (non-volatile) storage media**

Mass-Storage Management

- Usually disks used to store
 - data that does not fit in main memory or
 - **data that must be kept for a “long” period of time**
- Entire speed of computer operation hinges on disk subsystem and its algorithms
- OS activities
 - Free-space management
 - Storage allocation
 - **Disk scheduling**

Mass-Storage Management

- Some storage need not be fast
 - Tertiary storage includes optical storage, magnetic tape
 - Still must be managed – by OS or applications
 - Varies between WORM (write-once, read-many-times) and RW (read-write)

Performance of Various Levels of Storage

Level	1	2	3	4	5
Name	registers	cache	main memory	solid state disk	magnetic disk
Typical size	< 1 KB	< 16MB	< 64GB	< 1 TB	< 10 TB
Implementation technology	custom memory with multiple ports CMOS	on-chip or off-chip CMOS SRAM	CMOS SRAM	flash memory	magnetic disk
Access time (ns)	0.25 - 0.5	0.5 - 25	80 - 250	25,000 - 50,000	5,000,000
Bandwidth (MB/sec)	20,000 - 100,000	5,000 - 10,000	1,000 - 5,000	500	20 - 150
Managed by	compiler	hardware	operating system	operating system	operating system
Backed by	cache	main memory	disk	disk	disk or tape

Movement between levels of storage hierarchy can be explicit or implicit

Protection and Security



- **Protection** – any mechanism for controlling access of processes or users to resources defined by the OS
- **Security** – defense of the system against internal and external attacks
 - Huge range, including
 - denial-of-service,
 - worms,
 - viruses,
 - identity theft,
 - theft of service

Protection and Security

- Systems generally first distinguish among users, to determine who can do what
 - User identities (**user IDs**, security IDs) include name and associated number, one per user
 - User ID then associated with all files, processes of that user to determine access control
 - Group identifier (**group ID**) allows set of users to be defined and controls managed, then also associated with each process, file
 - Privilege escalation allows user to change to effective ID with more rights

Protection and Security

- A mechanism that ensures that the files, memory segments, CPU and other resources
- can be operated on by only those processes that have gained proper authorization from the OS.

Operating System Functions/Services

- One set of operating-system services **provides functionalities** that are helpful to the user:
 - **User interface** - Almost all operating systems have a user interface (**UI**).
 - Varies between **Command-Line (CLI)**, **Graphics User Interface (GUI)**, **Batch**
 - **Program execution** - The system must be able to load a program into memory and to run that program, end execution, either normally or abnormally (indicating error)
 - **I/O operations** - A running program may require I/O, which may involve a file or an I/O device

Operating System Functions/Services (Cont.)

- One set of operating-system services provides functions that are helpful to the user (Cont.):

File-system manipulation –

Programs need to

- read and write files and directories,
- create and delete them,
- search them,
- list file Information,
- permission management.

Operating System Functions/Services (Cont.)

- One set of operating-system services provides functions that are helpful to the user (Cont.):

Communications –

- Processes may exchange information,
 - on the same computer or
 - between computers over a network
- Communications may be via
 - shared memory or
 - through message passing (packets moved by the OS)

Operating System Functions/Services (Cont.)

- One set of operating-system services provides functions that are helpful to the user (Cont.):
 - **Error detection** – OS needs to be constantly aware of possible errors
 - May occur in the CPU and memory hardware, in I/O devices, in user program
 - **For each type of error, OS should take the appropriate action** to ensure correct and consistent computing
 - **Debugging facilities**

Operating System Functions/Services (Cont.)

- Another set of OS functions exists for **ensuring the efficient operation of the system itself via resource sharing**
 - Resource allocation
 - Accounting
 - Protection and security

Operating System Functions/Services (Cont.)

Resource allocation –

- When multiple users or multiple jobs running concurrently, resources must be allocated to each of them
 - Many types of resources -
 - CPU cycles,
 - main memory,
 - file storage,
 - I/O devices.
 - Some have
 - special allocation codes(CPU cycles, main memory, file storage) while
 - others (such as I/O devices) may have much more general request and release code.

Operating System Functions/Services (Cont.)

Accounting –

- To keep track of
 - which users
 - use how much and
 - what kinds of computer resources
- This record can be used for billing purposes or
 - for simply accumulating usage statistics.

Operating System Functions/Services (Cont.)

Protection and security –

- **Protection** involves ensuring that all access to system resources is controlled
- **Security** of the system from outsiders is also important, requires
 - user authentication,
 - extends to defending external I/O devices from invalid access attempts,
 - recording all such connections for detection of break ins.